

Roll No :

000 End Semester Examination 2024

Name of the Program: B. Tech

Semester: vii

Name of the Course: Deep Learning for Foundation

Course Code: TCS 783

Time: 180 Mins

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q 1)

(2X10=20 Marks) (CO2, CO3, CO5)

- (a) Suppose you train the one node neural network with the following parametric model where $Y = \{0, 1\}$ is Boolean variable. Suppose $w_0 = 0$, $w_1 = 1$, $w_2 = 1$.

$$P(Y = 1 | (x_1, x_2)) = \frac{1}{1 + \exp(w_0 + w_1 x_1 + w_2 x_2)}$$

Classify the given points according to the given parametric model of logistic regression.

- i) $(x_1 = 0, x_2 = 0)$
- ii) $(x_1 = 2, x_2 = -1)$
- iii) $(x_1 = -2, x_2 = 1)$
- iv) $(x_1 = 3, x_2 = -3)$

- (b) Explain how dropout works as a regularization technique and its effect on model performance.
- (c) You are tasked with designing a neural network for image classification. Explain how you would determine the architecture, activation functions, and training process.
- Q 2) (2X10=20 Marks) (CO1, CO2, CO4)
- (a) Discuss how does batch normalization help in improving the performance of networks.
- (b) Explain the role of convolution, pooling, and padding in CNNs. Provide examples of their applications.
- (c) Discuss how object localization differs from object detection with the help of the example.

Q 3)

(2X10=20 Marks) (CO3, CO4, CO5)

- (a) How residual network helps in training the very deep neural network with the help of the block diagram.
- (b) Explain the concept of word embeddings and their importance in natural language processing.
- (c) How does transfer learning enhance the performance of CNNs in specialized tasks such as object detection?

Q 4)

(2X10=20 Marks) (CO2, CO4, CO5)

- (a) What are the key differences between batch gradient descent, stochastic gradient descent, and mini-batch gradient descent?
- (b) Deep learning models often face challenges like vanishing gradients and slow convergence. Discuss how techniques like gradient clipping, learning rate scheduling, and weight initialization help address these challenges. Provide mathematical explanations where applicable.
- (c) Define deep learning and explain how it differs from traditional machine learning techniques.

Q 5)

(2X10=20 Marks) (CO1, CO3, CO4, CO5)

- (a) Develop a deep learning pipeline for medical image analysis. How would you handle limited labeled data using transfer learning?
- (b) Why are non-linear activation functions such as ReLU preferred over linear activation functions in deep neural networks?
- (c) Discuss the architecture and applications of autoencoder models in deep learning.